

NVDLA on PetaLinux 2024.1

Correctness, latency, throughput, and monitored energy at a glance

100 / 100

LeNet stability

One boot; no module reload or PL reset

246 / 246

ResNet-50 hardware layers

Exact match to source-built nv_small VP golden

1.34x

CPU INT8 execution advantage

378.7 ms CPU INT8 vs 507.7 ms NVDLA on ResNet-50

1.92x

NVDLA cold-deployment advantage

1.336 s NVDLA vs 2.566 s CPU INT8 on ResNet-50

-62.3%

NVDLA incremental energy

0.134 J NVDLA vs 0.356 J CPU INT8 on ResNet-50

20 x 2

Input-sensitivity control

Balanced image sets; three fresh boots per model

Comparison boundary: nv_small NVDLA INT8 is compared with both FP32 and independently quantized QDQ INT8 ONNX Runtime on four Cortex-A53 cores. Equal nominal precision does not imply identical quantization or graph transformation. The primary campaign contains 60 sessions; six supplementary sessions test 20 images per model.

Correctness before performance

Layered acceptance

OK

Pinned inputs

OK

ABI and build

OK

Verified nv_small
VP

OK

Driver probe

OK

GEM mapping

OK

IRQ + engine
completion

OK

Exact tensor +
repeat

LeNet / MNIST

Fast repeat oracle

INPUT
1 x 1 x 28 x 28LOADABLE
0.446 MBHARDWARE LAYERS
10ENGINE MIX
4 Conv / 4 SDP / 2 PDPVP
Exact outputBOARD
Exact; 100/100 repeats

Output: 0 2 0 0 0 0 0 124 0 0

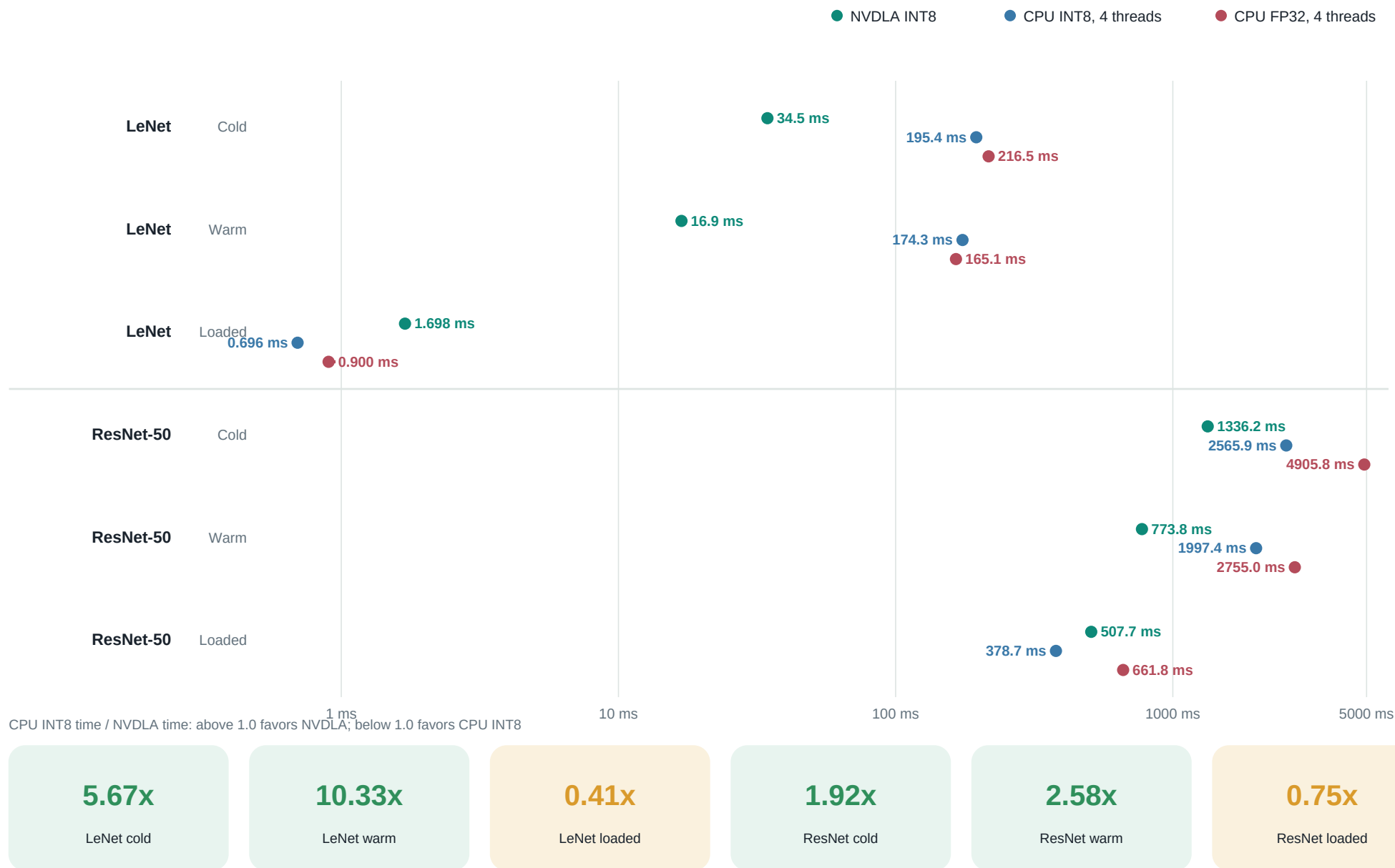
ResNet-50

Large multi-engine graph

INPUT
1 x 3 x 224 x 224LOADABLE
25.77 MBHARDWARE LAYERS
246ENGINE MIX
114 Conv / 130 SDP / 2 PDPVP
246/246; golden establishedBOARD
246/246; exact hash

Output: 1,000 signed values; SHA-256 842d34f...

Three deployed stacks across each timing boundary



Does execution time depend on the image?

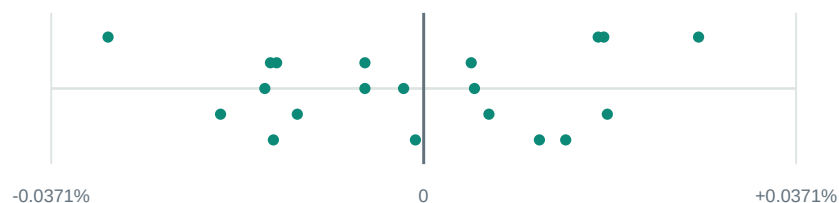
Execution-validity result: every input produced a repeat-stable output, increased IRQ activity, and completed without a bad kernel pattern.

LeNet / MNIST

20 inputs x 30 measured executions; loaded model and buffers reused

Per-input median execution deviation

model-specific horizontal scale



Observed range across the 20 medians: 0.0589%

Input decode and file I/O are outside this execution interval.

1.699 ms

Median execution

0.002 ms

Prepared input copy

20 / 20

Stable outputs

16 / 20

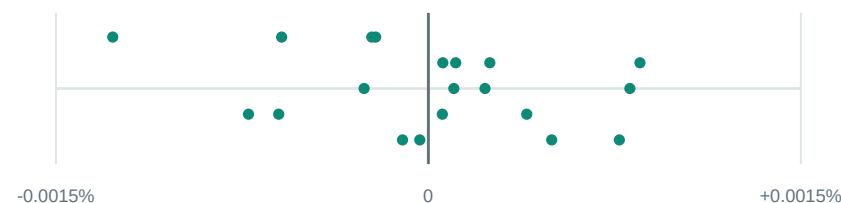
Recorded top-1

ResNet-50 / Imagenette

20 inputs x 15 measured executions; loaded model and buffers reused

Per-input median execution deviation

model-specific horizontal scale



Observed range across the 20 medians: 0.0022%

Input decode and file I/O are outside this execution interval.

507.8 ms

Median execution

0.171 ms

Prepared input copy

20 / 20

Stable outputs

17 / 20

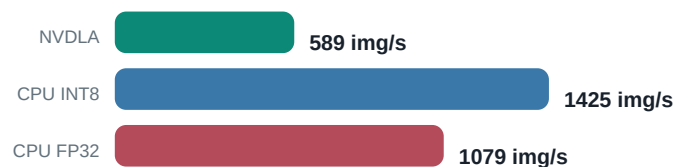
Recorded top-5

Interpretation: class accuracy is descriptive model-quality context from a small curated set. Hardware acceptance is based on repeat-stable output, IRQ progress, and clean kernel logs.

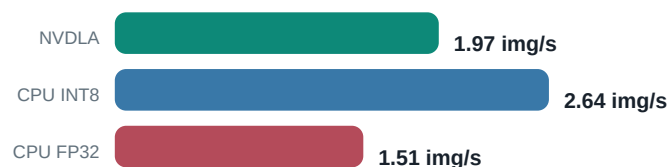
What changes between LeNet and ResNet-50?

Loaded-context throughput

LeNet



ResNet-50



Workload scale and operation mix

LeNet

NVDLA: 0.446 MB loadable | 10 hardware layers

CPU ONNX: 1.73 MB FP32 / 0.45 MB INT8



ResNet-50

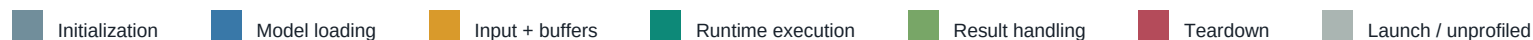
NVDLA: 25.77 MB loadable | 246 hardware layers

CPU ONNX: 102.48 MB FP32 / 26.09 MB INT8



ResNet-50 has **24.6x** more hardware layers and a **57.8x** larger loadable. Its larger execution graph amortizes fixed submission and interrupt costs.

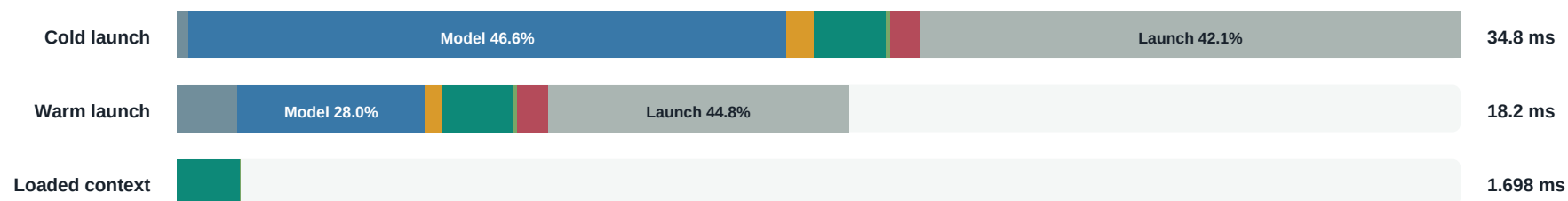
Where does NVDLA deployment time go?



LeNet

bar length is relative to this model's cold launch-to-exit time

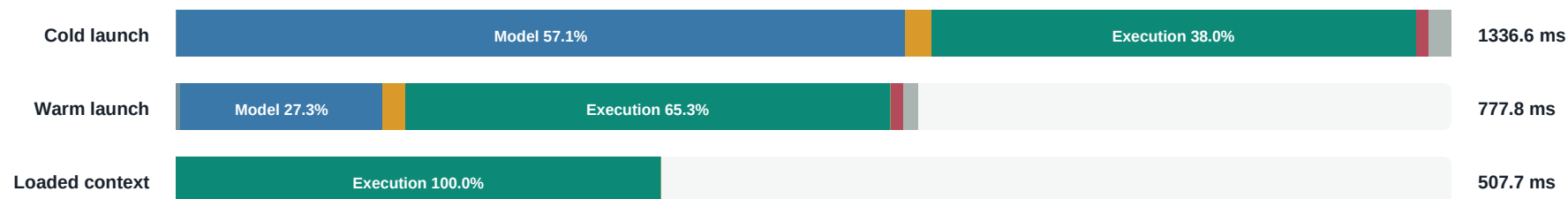
Warm runtime execution share 10.7% | loaded context 1.698 ms



ResNet-50

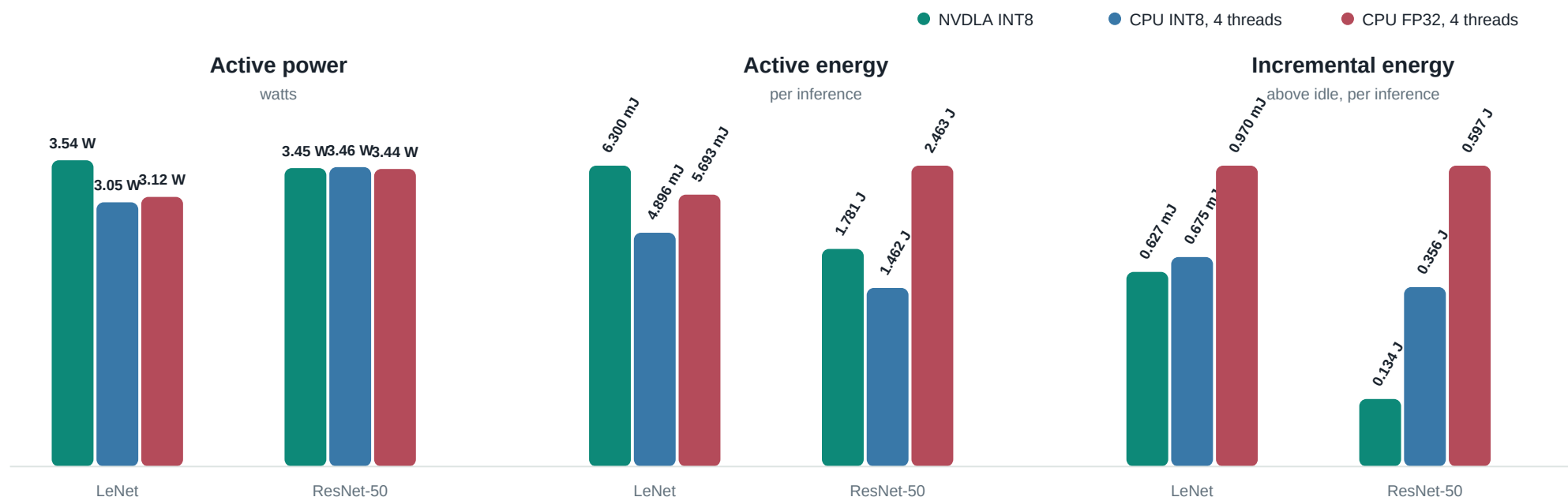
bar length is relative to this model's cold launch-to-exit time

Warm runtime execution share 65.3% | loaded context 507.7 ms



Timing boundary: cold and warm bars span process launch to exit. Loaded context reuses the model and buffers; its blocking runtime execution includes userspace dispatch, ioctl/KMD scheduling, accelerator work, IRQ completion, and emulator tasks.

Monitored PS + PL rails during inference



LeNet

+28.7%

NVDLA active energy

-7.1% incremental

Change is NVDLA relative to CPU INT8. Active includes the monitored platform; incremental subtracts driver-loaded idle.

ResNet-50

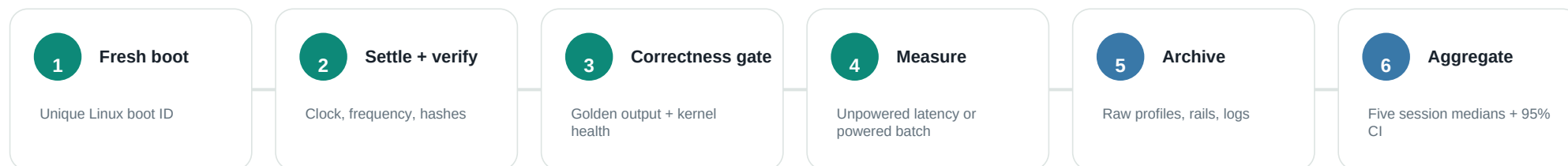
+21.8%

NVDLA active energy

-62.3% incremental

Change is NVDLA relative to CPU INT8. Active includes the monitored platform; incremental subtracts driver-loaded idle.

What was collected, and how to read it



Metric inventory

Correctness	Output hashes, exact tensors, engine sequence, IRQ deltas, repeat failures	Workload	Input shape, loadable/model size, HWL count, per-engine operation count
Latency	Cold deployment, warm deployment, loaded-context execution, runtime phases	Throughput	Images/s for every timing boundary; CPU/NVDLA latency ratios
Power	18 PS + PL rails, idle/active/incremental watts, integrated energy/inference	Environment	Kernel, binary hashes, clock, CPU affinity/frequency/governor, boot ID
Statistics	Raw samples, mean/median, spread, percentiles, retained outliers, bootstrap CI	Evidence	Serial, dmesg, runtime profiles, sensor samples, manifests, selection hashes

Interpret with: one ZCU102 and nv_small implementation; NVDLA INT8 versus independently quantized CPU INT8 plus an FP32 reference; monitored rails rather than wall input; five boots per primary cohort and three per input-sensitivity model.